Alien Technologies

Extraterrestrial Life, Exoplanets, and Alien Technologies

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PREFACE

Before You Begin

This book shows aliens only at the margins. We don't invent any: no names, no language, no culture, and the book won't end by predicting what they look like. The images in it come from film, books, comics, manga, and anime, and they're as big and strange and magnificent as people have imagined them. The movies turn up a little more often than the rest, but as teaching material, not for the thrill.

What this book shows instead is the real arithmetic. It asks what you can know about alien technology before you've ever seen any, and it answers with the one thing that's the same on every one of the countless worlds in our sprawling universe: physics, which is everywhere. Along the way it explains the extraterrestrial version of that physics, step by step, without formulas, without shortcuts. Some chapters are heavy going. We say so up front, so that nobody who needs a break in the middle thinks less of themselves for it.

The book has four parts, and it's only fair to announce them. The first collects what we humans have imagined about other forms of life over the decades and measures it against what we know and can prove. The second part tells a story: an archivist who finds something in old data, at night. The story is invented, and we say so here so that nobody finds out only at the end; but it's the most probable shape of a dream that many people share. First contact with alien life from somewhere beyond Earth will be fairly unspectacular. The story is written so that everything in it goes the way one would expect.

The third part checks the physics, chapter by chapter—energy, audibility, time, material, intent, chemistry, blueprint, circumstances, workshop—and it's the hardest part, the heavy going. The fourth turns the telescope around and says what follows from all of it: a thesis about who's transmitting out there, assuming anyone is transmitting at all. Most of our theses won't please anyone expecting visitors on Earth. We still think they're the most probable, and at every point we say how sure we are.

Three lines run through the book. The first is pure natural science; it lives in the running text. The second is the archivist; her name is Annie, and you'll recognize her chapters because they tell a story. The third runs along the margin. We call it the Overtone, like an overtone above the fundamental of the text: in almost every chapter there's a beautiful fantasy from science fiction or the history of science, with a short caption that says what possibilities it shows and which law it would actually founder on. That's where you, the reader, get what you may have been expecting and what the body text denies you—the aliens, in full splendor. Take them with you. The captions do take the dream away again, but only so that the real connections, and the contrasts with human imagination, come into view.

Two more things. First: all the numbers in this book come in two kinds—the measured ones, which we back up, and the estimated ones, which we also label as our estimate. We don't confuse the two, and we ask you not to either while reading. Second: we write "we," and by that we mean two authors, one of whom is human and one of whom isn't. That's stated in the imprint, and it belongs here too, because a book about technospheres shouldn't pretend it came into being without one.

It's now ten past three, and in the basement of an institute, a list has appeared …

3.1

Energy Coupling

The Question That Contains All the Others

Everything that lives hangs on a gradient, and everything that builds hangs on a bigger one. That's the second law of thermodynamics, which we'll come back to at leisure; here we're concerned with its practical side. A living thing is a channel through which energy flows from a high level to a low one, and what it siphons off along the way is its life. A technosphere is a second channel, wider, running alongside the first and at some point overtaking it. The question of what both hang on—which gradient, how wide, for how long—decides everything else: how big they get, how far they travel, whether they transmit, whether anyone sees them.

And it decides what's left of the imagination. Because every technology the movies have ever shown comes with a bill, and that bill appears in no screenplay. This chapter settles it. It is, admittedly, the chapter in which most of the spaceships die.

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The Staircase

Energy sits in matter on three steps, and between the steps lie not factors but chasms.

The first step is chemistry. Whoever breaks bonds and forms new ones—burns sugar, burns gasoline, combines hydrogen with oxygen—gets about

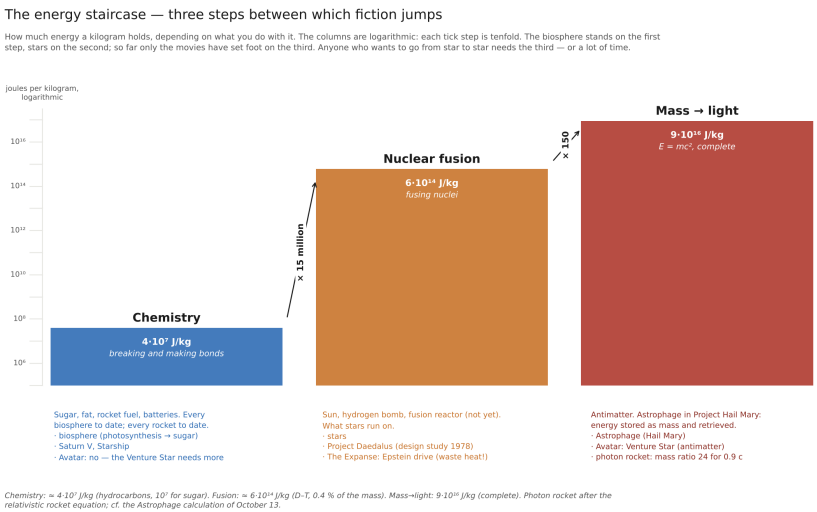


Figure 11 · The energy staircase—three steps, between which the imagination jumps

forty million joules out of a kilogram. A lot, if you're a car, and it's all a biosphere has ever had: photosynthesis, respiration, muscles, fire, every rocket that ever launched. Earth has lived on the first step for four billion years, and it lives well on it.

The second step is nuclear fusion. Whoever fuses hydrogen nuclei into helium gets six hundred trillion joules out of a kilogram—fifteen million times more than from chemistry. The stars stand on this step, and Earth's hybrid technosphere has been working on it for seventy years, with a result politely described as "under construction." Whoever sets foot on it has energy no chemistry can ever deliver, and a problem no chemistry ever had: the immense waste heat has to go somewhere.

The third step is the complete conversion of mass into energy, E equals mc^2, ninety quadrillion joules per kilogram, a hundred and fifty times more than fusion. It exists in physics—matter and antimatter annihilating each other do exactly that—and it exists in no workshop. Producing antimatter costs more than it gives back, and storing it is an unsolved problem, because it dissolves

every container you offer it. So far, only the movies have set foot on the third step.

That's the whole staircase, and what sets it apart from every other staircase is that you don't climb it in steps. Between chemistry and fusion lies a factor of fifteen million; between fusion and matter, a factor of a hundred and fifty. Whoever stands on the first step can't do a little bit of fusion. He can master it or not, and as long as he can't, everything he does is chemistry—even when it's very big.

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What Stars Hang On, and What We Do

Stars hang on the second step, and they hang well: the sun burns six hundred million tons of hydrogen per second and radiates the energy in all directions, and of what it radiates, Earth catches one two-billionth, around a hundred and seventy thousand terawatts; after subtracting what clouds and ice throw back, a hundred and twenty thousand remain. Everything that lives on Earth lives off this two-billionth—and in fact off a tenth of it, because the plants don't catch more than that, and off one percent of that tenth, because they don't turn more than that into sugar.

Earth's technosphere to date turns over around twenty terawatts, one six-thousandth of what stays at the ground. For the most part it draws this not from the light that arrives, but from the light that arrived a hundred to three hundred million years ago and stayed underground as coal and oil. In the terms of this chapter, that's a technosphere hanging on a stockpile, not on a gradient. Stockpiles run out. Gradients don't, as long as the star keeps shining.

That's why we speak of energy coupling and not of energy consumption. The question to ask a technosphere isn't how much it consumes, but what it hangs on—a stockpile that ends in the short term, or a star that won't end for a very much longer time. And coupling to the star is the transition every technosphere either makes or doesn't. Ours is attempting it right now, the

turn away from fire, with solar cells as direct energy use and fusion reactors that promise the second step. Nikolai Kardashev proposed a scale for this in 1964 that everybody knows and nobody likes: Type I uses the energy of its planet, Type II that of its star, Type III that of its galaxy. On this scale we're about 0.7. That sounds like a grade. It's an address: our energy supply doesn't hang on our star yet.

A biosphere always depends on its own star—that's its advantage—but it's highly inefficient: it harvests only a thousandth of the light, and it stores the harvest as sugar and fat, on the first step, chemistry. From this follows the sentence that carries this chapter and that has never been spoken in a movie: biology harvests and stores badly. Whoever wants to go from star to star needs a kind of storage no cell has.

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The Miracle, Thought Through to the End

There's one film that starts at exactly this point and thinks it through to the end, which is why it's the best case for this chapter: *Project Hail Mary*, in theaters in 2026, after Andy Weir's novel of 2021.

The Astrophage are microbes that live in starlight and store energy in a way no earthly cell can: not as sugar, but as mass. They take in light and get heavier; they give off the energy and get lighter, as light of a single wavelength in the infrared, the Petrova line. The third step, in other words, inside a living thing. And Weir, unlike almost every other author, did the math for it.

If a spaceship is to fly on Astrophage, then as a photon rocket: it emits light, and light has the highest exhaust velocity there is. The rocket equation then tells you how much fuel you need to reach a given speed and brake again. For ninety-two percent of the speed of light, accelerating and braking again, it yields a mass ratio of about twenty-four: for every ton of ship that arrives, twenty-three tons of Astrophage have to be on board at launch. Weir's ship weighs about two thousand tons at launch and just under ninety at the end, and that fits. And that's only the outbound leg; that's why Weir's flight is a

one-way trip. The film holds up, except in one place: Astrophage can't exist, because nothing made of atoms stores energy at this density as mass without exploding. It's a miracle, thought through to the end—and that makes it the only spaceship in this book that doesn't die of the math, only of the premise.

What the Astrophage hang on is the light of the sun, as directly as plants do. They're a biosphere that takes the star's entire yield for itself—and that's why, in the logic of this book, they're the only thing the movies have ever shown as a biosphere that behaves like a technosphere. Rocky, the other alien in this film, isn't a microbe but an engineer, and he flies on the same fuel because, like us, he has no better one. Two civilizations, sixteen light-years apart, both needing the same miracle to meet: that's the most honest statement about interstellar flight ever put on film. Without that miracle, nobody would ever meet anybody.

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Three Spaceships and Their Bill

The other spaceships from the big pictures have no such miracle, and so they die here, one after another, in three sentences.

Whoever flies, shines. The Venture Star in *Avatar* is the most honest ship of the imagination after Weir's: it flies on antimatter and a laser sail, takes six years to Alpha Centauri, and has radiator panels the size of a football field at its stern, because the drive produces waste heat. All of that is correct, and it leads to a conclusion the film doesn't follow through: a drive that brings a ship to seventy percent of the speed of light radiates a hundred thousand times what the entire technosphere of Earth turns over, and its exhaust plume is visible across light-years. Every ship flying interstellar would be a beacon. You'd see it before it arrives—and you'd see it everywhere you looked, if ships really were flying somewhere. The fact that you see nothing is an observation too.

Whoever hovers, pays. The twelve ships in *Arrival* hang motionless in the air, without propulsion, without noise, without heat. Keeping something in the air without support and without buoyancy costs power, every second, and

at least as much as it costs to lift it; a helicopter holding position burns through its tank doing so. A ship the size of a skyscraper hovering over Montana turns over a power plant's worth, and the energy goes somewhere—as heat, as wind, as radiation. In *Arrival* it goes nowhere. That's the break point, and it's the same one as in *District 9*, where the mothership hangs over Johannesburg for twenty-eight years without anyone asking what on. Whoever hovers, pays. Whoever doesn't pay, doesn't hover.

Whoever travels between stars needs no planet. The Tet in *Oblivion* sucks up Earth's oceans to make energy from them, and that's like someone pulling up to a bicycle pump to refuel a private jet. Whoever can travel across stars has set foot on the third step, or at least the second, and on those steps a planet is no longer a stockpile but a speck of dust. The gradient the journey cost is bigger than anything you could harvest at the destination. That sentence disposes in one line of every invasion ever filmed, from *The War of the Worlds* to *Independence Day*: whoever can come doesn't have to. Whoever has to come can't.

And *The Expanse*, which tried hardest, dies most honestly: the Epstein drive, a fusion engine that accelerates ships at one g for weeks, fails not on thrust but on waste heat. An engine turning over that much power would need radiators bigger than the ship, and the series didn't draw them because they'd be ugly. It's the most polite way to fail: on the aesthetics of thermodynamics.

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The Trace Nobody Avoids

From all this follows something that matters more for the search than for the movies. Energy that gets turned over ends up as heat, and heat radiates. A technosphere using one percent of its star radiates one percent of its star in the infrared, as heat at a few hundred kelvin, and that can be seen—across hundreds of light-years, with instruments that have been in operation since the eighties. Freeman Dyson proposed searching for it in 1960, and Project

Hephaistos has been doing so since 2022, with the catalogs of Gaia and WISE: millions of stars, checked for heat that doesn't fit the spectrum.

It found candidates, seven, and for five of them the excitement was brief: three turned out in the radio to be distant dusty galaxies, and for candidates D and E the James Webb telescope in 2026 showed a galaxy apiece, glowing in the infrared and happening to line up with the star. That's where things stand: there's no technosphere within sight that wraps up a noteworthy part of its star, a tenth or more. A strong statement, and a negative one, which is why it's rarely put this way. It reads: whoever is building out there is building small. Or cold. Or not at all.

And it leads straight to Annie's object. On the twelfth night she checked the location of the source in the infrared, with the same logic—is there a body there, a galaxy behind it?—and found nothing. That's the point where physics takes her find seriously for the first time, precisely because nothing is there. Something that's been transmitting for twenty-six years turns over power, and power is heat. A transmitter still measurable at a hundred light-years needs at least the power of a power plant if it beams, and that of an entire world if it radiates in all directions. Both are warm—but both are invisible from a hundred light-years, a power plant as much as a world; what you see at that distance in the infrared are stars, brown dwarfs, young giant planets. So the fact that none of those is there says nothing about the power, but a great deal about the body: whatever is transmitting next to the red dwarf is no planet and no star. It's small. That's no proof. It's the absence of a contradiction, and in this search through the possibilities, that's a lot.

From a Heavy World. There's one question that sharpens all the energy considerations of this chapter into a single question: How do you get away at all? On Earth a rocket needs a good nine kilometers per second for orbit and a good eleven to escape gravity, and with hydrogen and oxygen—the best chemistry there is for the purpose—that means ninety percent of the launch mass is fuel.

Tight, but it works; Earth sits at the edge of the feasible. Super-Earths, the most common rocky worlds in the galaxy, lie beyond it. Michael Hippke calculated in 2018 that for a world of ten Earth masses the escape velocity is twenty-seven kilometers per second and the rocket for an Apollo capsule would have to weigh four hundred thousand tons—a hundred and thirty Saturn V rockets stacked on top of each other. Chemical spaceflight isn't impossible there, but unaffordable; the limit lies at about ten Earth masses, and it gets expensive starting at two or three Earth masses. Whoever wants to leave from there needs catapults (energy stays on the ground), beamed propulsion (energy gets sent after you), or nuclear pulses (the second step, very brutal). A space elevator doesn't help: on a heavy world the cable would have to be made of a material stronger than anything chemistry offers—nanotubes are barely enough for Earth—and somebody has to get the first cable up there in the first place. The threshold is a one-time thing—whoever gets up once builds skyhooks and orbital rings and climbs cheaply after that—but it's fairly high. The consequence for the search: most worlds where something is building are worlds that are hard to get away from. They'll transmit because they can't fly.

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One Page of Step Three

This book stays on the lower two steps, because that's where the physics we know is in charge. But it would be dishonest not to at least look at the third, because the imagination lives there, and the question of what a technosphere does once it has set foot on the third step isn't nonsensical. It's just unanswerable, and it's worth knowing why.

A technosphere that converts mass completely into energy no longer has an energy problem. It does have a waste-heat problem, though, because everything it turns over has to be radiated away, and the coldest way to radiate

heat is to spread it out wide and slow: a shell around the star, glowing at a few degrees above absolute zero. That's not the Dyson sphere of the imagination, which would be warm, but its limiting case—a thing so well cooled that it's barely distinguishable from the background of the universe. Freeman Dyson saw that too; it's the reason the search for warm technospheres finds only the young and the loud.

And that leads to a thought we'll set down here and not pursue, because it leads into theology if you follow it to the end: a step that lives entirely within physics is indistinguishable from physics. Its signal looks like noise, because compression is noise; its structures look like stars, because well-cooled structures look that way; its shell would lie around everything and from the inside wouldn't be recognizable as a counterpart, only as surroundings. The question "Are we alone?" would then be wrongly posed. We wouldn't be alone. We'd be inside.

That's a beautiful thought, and it's dangerous for exactly the reason that it explains everything: every silence, every regularity, every constant could be read as a message, and what explains everything explains nothing. We show the step, let it stand, and add that the same observations can be explained without a shell. A pattern with two points is a long way from being a curve.

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What's Left of the Imagination

After this chapter, energy carries its two markers—it isn't one of the seven axes from the foray, it lies underneath all seven, because whoever is big, thinks fast, or wants to be heard needs it first. Where we are: on the first step, with one foot on the second, hanging on a stockpile that's finite. Where the limits lie: at waste heat, on every step, and at the mass ratio, for every ship. In between is everything the movies have shown, and most of it lies beyond the limit.

What's left is little. It's ships that shine and would therefore be seen; technospheres that radiate heat and would therefore be found, or would be cold



OVERTONE

Dyson Sphere Program, 2021. One robot, some conveyor belts, and after a week of play a shell hangs around the star. — Timesheet: Kardashev Type II, done in a bit of overtime. — Remark from physics: The energy staircase has three steps, with factors of fifteen million and a hundred fifty between them. One thing the game booked correctly: it starts with a plain piece of iron.

and therefore not. It's worlds you can't get away from and that therefore are more likely to transmit. And an object in Field 2214 that's warm, transmits, and moves—and that fits this picture like nothing the imagination has ever drawn.

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. . .

What remains. Energy sits in matter on three steps—chemistry, fusion, mass—and between them lie factors of fifteen million and a hundred and fifty that you don't climb in steps. A biosphere harvests a thousandth of its star and stores on the first step; a technosphere is decided by whether it hangs on a stockpile or on the star. Every ship dies of the mass ratio or of waste heat, and whatever flies, shines. Whoever can come doesn't have to; whoever has to come can't. And because power is heat, every transmitter radiates—but from a hundred light-years, all you see in the infrared are stars and worlds. That nothing shines next to the red dwarf in Field 2214 says nothing about the power and a great deal about the body: it's small.

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That was the sample.

Excerpt from Part 3 · The Physics

Alien Technologies. Extraterrestrial Life, Exoplanets, and Alien Technologies. What if the first extraterrestrial encounter begins with a signal?

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